Quiz I

Test your knowledge of different systems that you have learned in this course by completing this quiz.

Quiz ?

Statement

A cache is primarily used to improve:

Select Multiple Options

- [ ] Latency
- [ ] Concurrency
- [ ] Fault tolerance
- [ ] Cost

Check Answer

Quiz ?

Statement

In a distributed environment, a network partition can cause which of the following problems? Select all that apply.

Select Multiple Options

- [ ] Split-brain
- [ ] Each of the two servers might think that the other one is dead.
- [ ] Messages might be sent to the wrong server.
- [ ] Message or data loss.

Check Answer

Quiz ?

Statement

What is the benefit of using Piggybacked acknowledgments?

Select Multiple Options

- [ ] It delays the acknowledgment response.
- [ ] It efficiently uses the available channel bandwidth, thus improving the efficiency of the system.
- [ ] It decreases the system complexity.
- [ ] It is an efficient way for the sender to acknowledge the receipt of an acknowledgment.

Check Answer

Quiz ?

Statement

The CAP theorem tells us that:

Select Multiple Options

- [ ] A distributed system can only have two out of three. Consistency, atomicity, and partition tolerance.
- [ ] A distributed transaction must have to choose two out the three. Consistency, isolation, and durability.
- [ ] Any distributed system which is highly-available will have problems with consistency.
- [ ] In distributed systems, durability and lower latency can only come at the cost of lower consistency.

Check Answer

Quiz ?

Statement

Following is the property of the Write-ahead Log (WAL). Select all that apply.

Select Multiple Options

- [ ] Data is always sequentially appended to the WAL.
- [ ] Each log entry in the WAL has a unique identifier.
- [ ] WAL does not allow updating the data once written.
- [ ] WAL reduces the read latency by providing a cache containing recent transactions.

Check Answer

Quiz ?

Statement

As opposed to locks, leases:

Select Multiple Options

- [ ] Have expiration times.
- [ ] Are long-term while locks are short-term
- [ ] Can be shared between multiple clients
- [ ] Are coarse-grained locks

Check Answer

Quiz ?

Statement

What advantage does it give to a system to have separate read and write locks?

Select Multiple Options

- [ ] Enables concurrent reading of a resource by multiple entities
- [ ] Enables concurrent writing to a resource by multiple entities
- [ ] Enables entities to get 'execute' access on a resource if they acquire both read and write locks at the same time
- [ ] None of the above

Check Answer

Quiz ?

Statement

For GFS, what are the benefits of a large chunk size? **Select all that apply.**

Select Multiple Options

- [ ] A small file can all be stored in just one chunk.
- [ ] Large chunk size provides highly efficient sequential reads and appends of large amounts of data.
- [ ] A large chunk size simplifies ChunkServer management.
- [ ] A large chunk size reduces the size of the metadata stored on the master.

Check Answer

Quiz ?

Statement

Consistent hashing ensures that:

Select Multiple Options

- [ ] Data will be stored on the servers reliably.
- [ ] If a server is removed, its data will be distributed to all other nodes in the cluster.
- [ ] Only a small set of keys move when servers are added or removed.
- [ ] The data will be distributed equally among the servers.

Check Answer

Quiz ?

Statement

Dynamo uses virtual nodes for which of the following reasons? Select all that apply.

Select Multiple Options

- [ ] To avoid hotspots
- [ ] To spread the load more evenly across the physical nodes on the cluster by dividing the hash ranges into smaller subranges
- [ ] To enable the use of virtual machines and virtual memory
- [ ] To better maintain a cluster containing heterogeneous machines

Check Answer

Quiz ?

Statement

Of the following options, which is not a characteristic of Amazon's Dynamo?

Select Multiple Options

- [ ] Partition-tolerant
- [ ] Highly available
- [ ] Strongly Consistent
- [ ] Highly reliable

Check Answer

Quiz ?

Statement

Unlike BigTable, with Amazon Dynamo:

Select Multiple Options

- [ ] Keys are sorted alphabetically to support iteration.
- [ ] An object is identified by exactly one key.
- [ ] Two processes may write conflicting updates.
- [ ] All requests pass through a coordinator.

Check Answer

Quiz ?

Statement

Virtual nodes in Amazon Dynamo are designed to:

Select Multiple Options

- [ ] Improve fault tolerance due to the replication of nodes
- [ ] Increase the requests the system can handle since many virtual nodes can be managed by one physical node
- [ ] Improve load distribution when adding or removing nodes
- [ ] Create an overlay network that arranges nodes into a logical ring

Check Answer

Quiz ?

Statement

Which of the following are in-memory structures where Cassandra buffers write?

Select Multiple Options

- [ ] SSTables
- [ ] NetTables
- [ ] LinkTables
- [ ] Memtables

Check Answer

Quiz ?

Statement

What are the disadvantages of using Tombstones in Cassandra? Select all that apply.

Select Multiple Options

- [ ] Slower reads
- [ ] Deleted data takes more storage until Tombstones are completely removed during compaction.
- [ ] Slower writes
- [ ] Faster writes

Check Answer